

Energy-Gated Frequency Networks

Supplementary Material

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This supplement preserves diagnostic figures removed from the compact main paper. They document the historical model progression but do not resolve the causal contribution of gating. Reported validation values are descriptive results from development experiments.

1 Filter-bank evolution

The hard synthetic task motivated a progression from five coarse fixed filters to eight fixed filters and then to free FIR kernels. The free kernels improved validation accuracy, but their spectral responses became irregular and multi-band. Consequently, initialized band names are not physical guarantees after unconstrained learning.

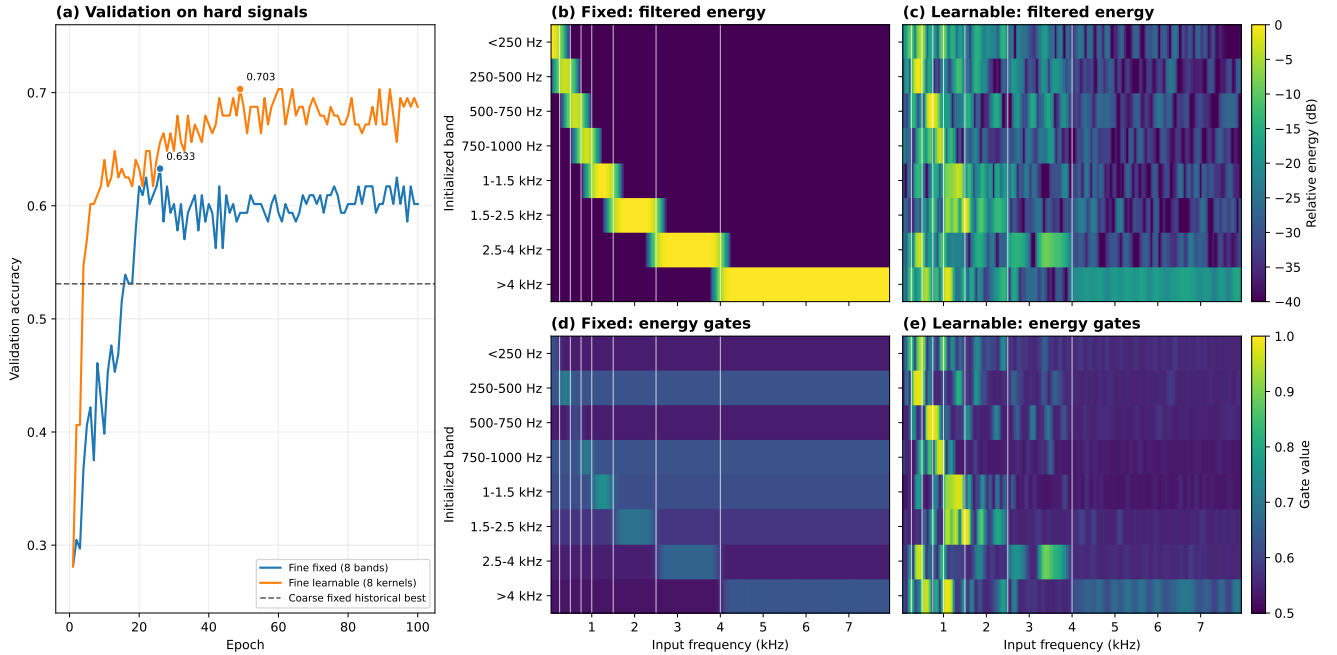


Figure 1: Evolution from coarse fixed filters to a fine fixed bank and free FIR kernels. Free kernels improve the hard-task result while weakening explicit low-pass, high-pass, and band-pass interpretation.

2 Gate variants

Independent, contextual, and sparsity-regularized gates were compared on the hard synthetic validation split. Nearly all gates remained above 0.5; a low mean activation or a threshold-dependent active-band count should therefore not be interpreted as proof of discrete routing.

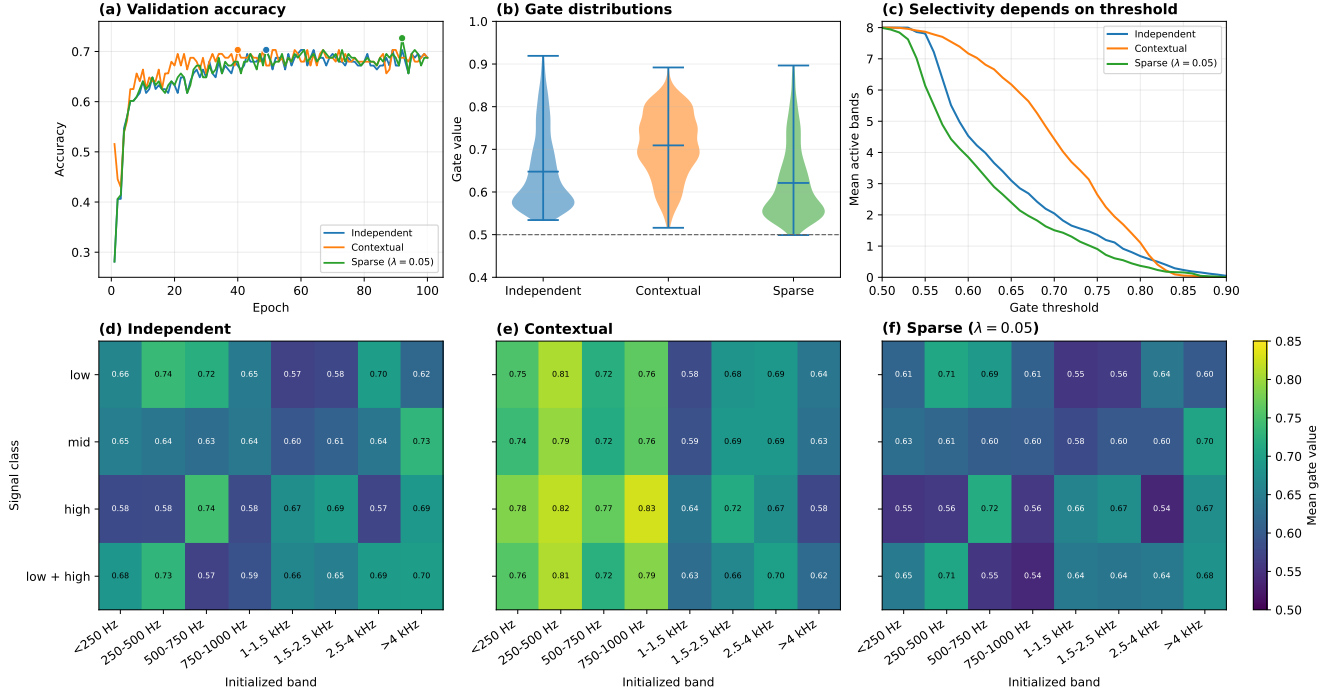


Figure 2: Independent, contextual, and sparsity-regularized gate behavior. The results support soft attenuation, not hard feature selection.

3 FSDD split sensitivity

A random recording split contains voices observed during training, whereas the speaker-disjoint protocol evaluates an unseen voice. The large performance gap shows that the random split cannot be used as evidence of speaker invariance.

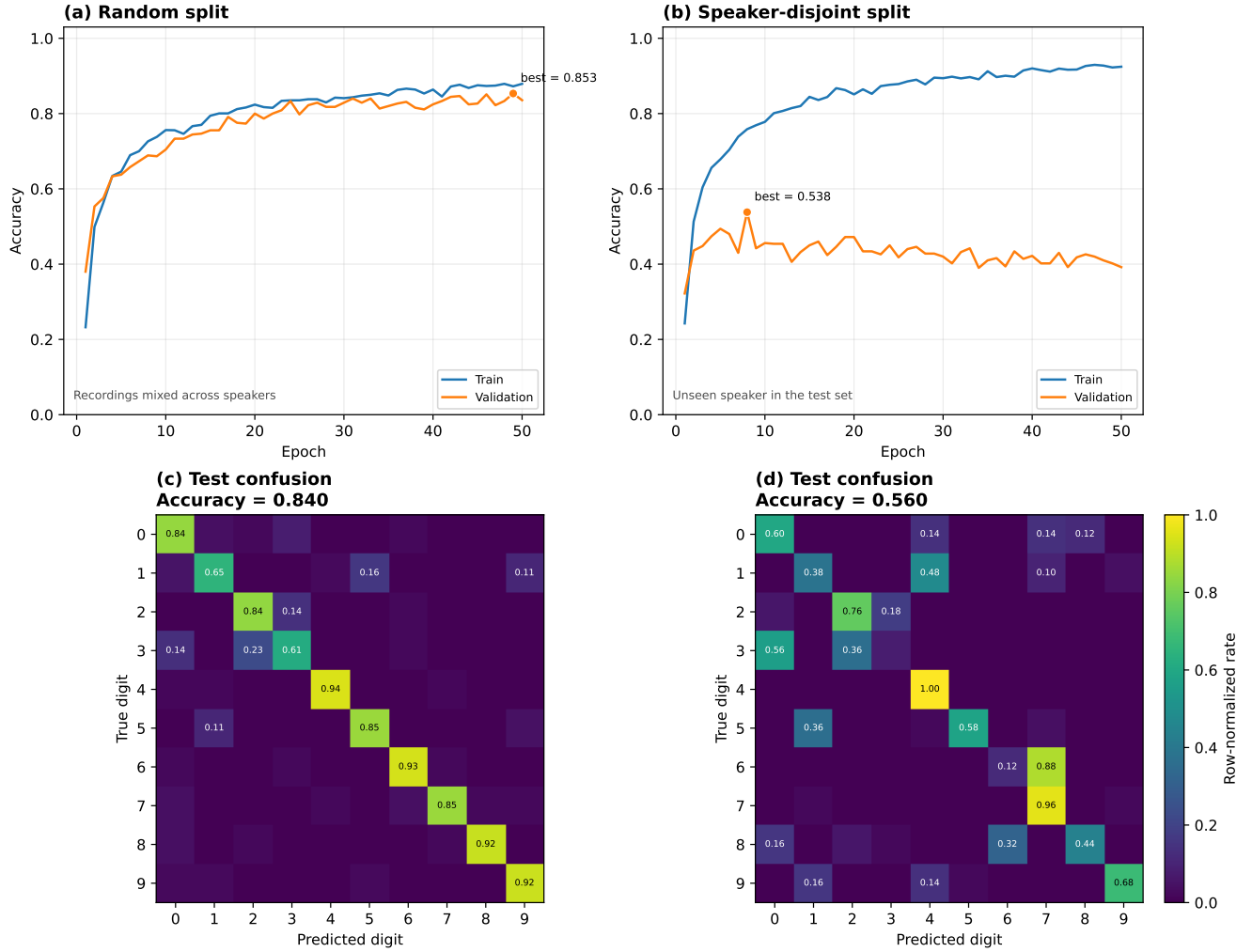


Figure 3: Random versus speaker-disjoint FSDD evaluation, including training histories and normalized confusion matrices.

4 FSDD temporal head comparison

Replacing global pooling with a temporal head recovered 14.8 percentage points on the speaker-disjoint test, but Conv1D retained the best test accuracy with fewer parameters than Temporal EGFN. This result supports the importance of temporal order rather than an isolated advantage of the frequency frontend.

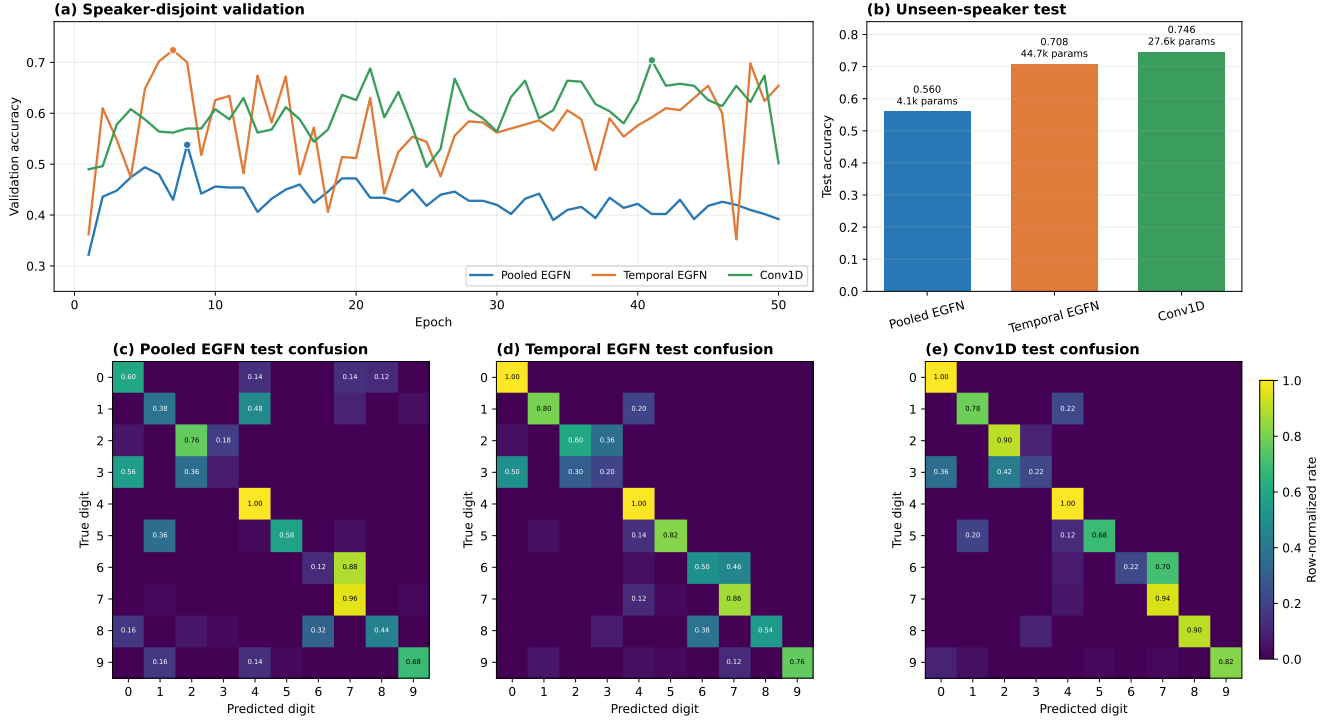


Figure 4: Pooled EGFN, Temporal EGFN, and Conv1D under speaker-disjoint FSDD evaluation.